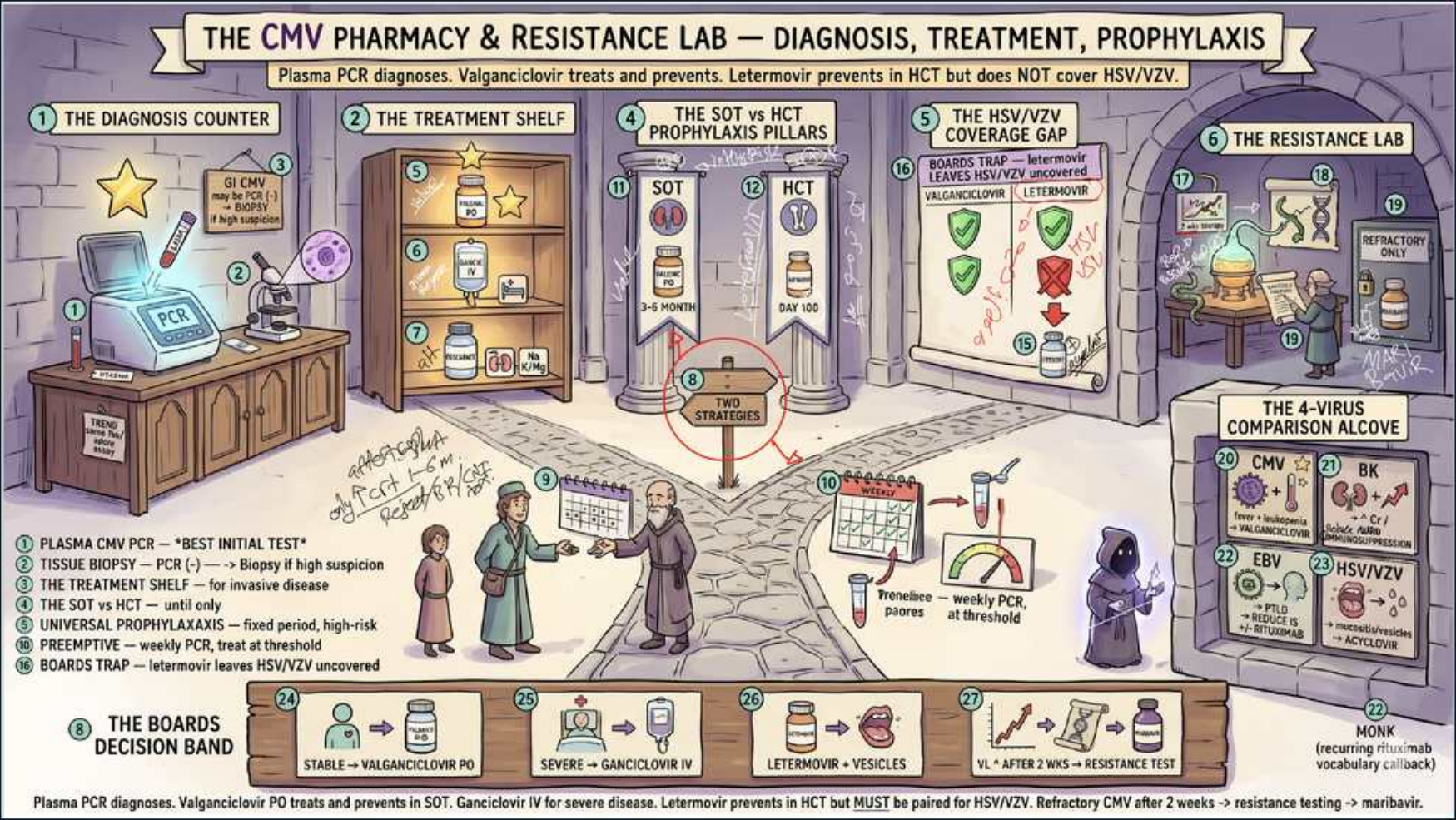


Post-Transplant CMV: Diagnosis, Treatment & Prophylaxis Pharmacology



1. Plasma CMV PCR — best initial test

WHAT YOU SEE

Diagnosis counter, PCR machine, 'BEST INITIAL TEST'

WHAT IT ENCODES

Quantitative plasma CMV PCR (viral load) is the best initial test to diagnose and monitor CMV in transplant recipients — it detects viremia, guides preemptive therapy, and tracks treatment response. Rising viral load signals active disease.

BOARD TRAP

Relying on CMV serology to diagnose active disease post-transplant — serology shows prior exposure; quantitative PCR detects active replication.

2. Tissue biopsy for invasive/GI CMV

WHAT YOU SEE

GI CMV PCR(–) → biopsy, 'invasive disease'

WHAT IT ENCODES

In suspected tissue-invasive CMV (e.g., colitis) plasma PCR may be negative; tissue biopsy showing characteristic owl-eye inclusions / immunostaining confirms invasive disease. Biopsy is required when PCR is negative but clinical suspicion is high.

BOARD TRAP

Excluding CMV colitis because plasma PCR is negative — tissue-invasive GI disease can have low/absent viremia; biopsy is needed.

3. Ganciclovir IV — severe disease

WHAT YOU SEE

SEVERE → GANCICLOVIR IV

WHAT IT ENCODES

IV ganciclovir is first-line for severe or tissue-invasive CMV disease. It is a guanosine analog activated by viral UL97 kinase, inhibiting CMV DNA polymerase. Main toxicity is bone-marrow suppression (neutropenia). Treat until viral clearance.

BOARD TRAP

Using oral valganciclovir for severe/invasive CMV — severe disease needs IV ganciclovir; oral valgan is for non-severe/maintenance.

4. Valganciclovir PO — stable disease

WHAT YOU SEE

STABLE → VALGANCICLOVIR PO

WHAT IT ENCODES

Oral valganciclovir (prodrug of ganciclovir) treats non-severe CMV disease and is the standard agent for prophylaxis. Same mechanism and marrow toxicity as ganciclovir. Treat until viral load is undetectable.

BOARD TRAP

Choosing valganciclovir for life-threatening/invasive CMV — reserve oral valgan for mild disease and prophylaxis; use IV ganciclovir if severe.

5. Preemptive vs universal prophylaxis

WHAT YOU SEE

SOT vs HCT 'TWO STRATEGIES' pillars

WHAT IT ENCODES

Two prevention strategies: universal prophylaxis (antiviral for all at-risk for a fixed period, e.g., D+/R– kidney) versus preemptive therapy (serial PCR monitoring, treat only when viral load crosses a threshold). High-risk D+/R– favors universal prophylaxis.

BOARD TRAP

Treating D+/R– (donor-pos, recipient-neg) with preemptive monitoring alone — highest-risk serostatus warrants universal prophylaxis.

6. Preemptive = weekly PCR, treat at threshold

WHAT YOU SEE

Calendar 'weekly PCR at threshold'

WHAT IT ENCODES

Preemptive therapy uses weekly quantitative CMV PCR; antiviral starts once viral load reaches a defined threshold, avoiding drug exposure in those who never reactivate. Requires reliable, frequent monitoring to catch early replication.

BOARD TRAP

Choosing preemptive strategy when reliable weekly PCR isn't feasible — without close monitoring, universal prophylaxis is safer.

7. Letermovir — HCT prophylaxis, no HSV/VZV

WHAT YOU SEE

HSV/VZV coverage gap shields, 'letermovir leaves HSV/VZV uncovered'

WHAT IT ENCODES

Letermovir inhibits the CMV terminase complex and is used for CMV prophylaxis (notably in HCT). It is CMV-specific — it does NOT cover HSV or VZV, so acyclovir/valacyclovir must be added for herpes prophylaxis.

BOARD TRAP

Assuming letermovir covers HSV/VZV — it is CMV-only; pair it with acyclovir/valacyclovir to cover herpes simplex and varicella.

8. Boards trap — letermovir leaves HSV/VZV uncovered

WHAT YOU SEE

BOARDS TRAP banner, 'letermovir leaves HSV/VZV uncovered'

WHAT IT ENCODES

Key board point: ganciclovir/valganciclovir cover CMV plus HSV/VZV, but letermovir's narrow CMV-terminase mechanism leaves a herpesvirus coverage gap. Recognize when a second agent is required for HSV/VZV prophylaxis.

BOARD TRAP

Stopping all acyclovir when switching to letermovir — letermovir's coverage gap leaves HSV/VZV unprotected.

9. Resistance — UL97 / UL54 mutations

WHAT YOU SEE

Resistance lab, REFRACTORY CMV

WHAT IT ENCODES

CMV refractory after ~2 weeks of appropriate therapy raises concern for resistance — UL97 mutations confer ganciclovir resistance, UL54 (DNA polymerase) confers broader resistance. Send genotypic resistance testing to guide salvage therapy.

BOARD TRAP

Continuing ganciclovir despite a rising viral load on treatment — refractory CMV at ~2 weeks needs resistance testing, not dose persistence.

10. Foscarnet — resistant CMV salvage

WHAT YOU SEE

Resistance alcove salvage vial

WHAT IT ENCODES

Foscarnet directly inhibits CMV DNA polymerase without requiring UL97 activation, so it treats UL97-mediated ganciclovir-resistant CMV. Major toxicity is nephrotoxicity and electrolyte wasting (hypocalcemia, hypomagnesemia, hypokalemia).

BOARD TRAP

Picking another guanosine analog for UL97-resistant CMV — foscarnet (no kinase activation needed) is the salvage; watch nephro/electrolyte toxicity.

11. 4-virus comparison — BK, EBV, HHV

WHAT YOU SEE

4-VIRUS COMPARISON ALCOVE: CMV, BK, EBV, HHV

WHAT IT ENCODES

Post-transplant DNA viruses behave differently: CMV (colitis/pneumonitis, PCR-guided antivirals), BK polyomavirus (nephropathy — reduce immunosuppression, no good antiviral), EBV (PTLD — reduce immunosuppression ± rituximab). Distinguish their management.

BOARD TRAP

Treating BK nephropathy with antivirals like CMV — BK is managed by lowering immunosuppression; antivirals don't reliably work.

12. Reduce immunosuppression as adjunct

WHAT YOU SEE

Treatment shelf with immunosuppression knobs

WHAT IT ENCODES

Reducing net immunosuppression is a key adjunct in CMV (and essential in BK/EBV PTLD) because over-immunosuppression drives viral reactivation. Balance against rejection risk while antiviral therapy controls active CMV.

BOARD TRAP

Maximizing antivirals while leaving heavy immunosuppression untouched — failure to reduce immunosuppression perpetuates viral reactivation.
